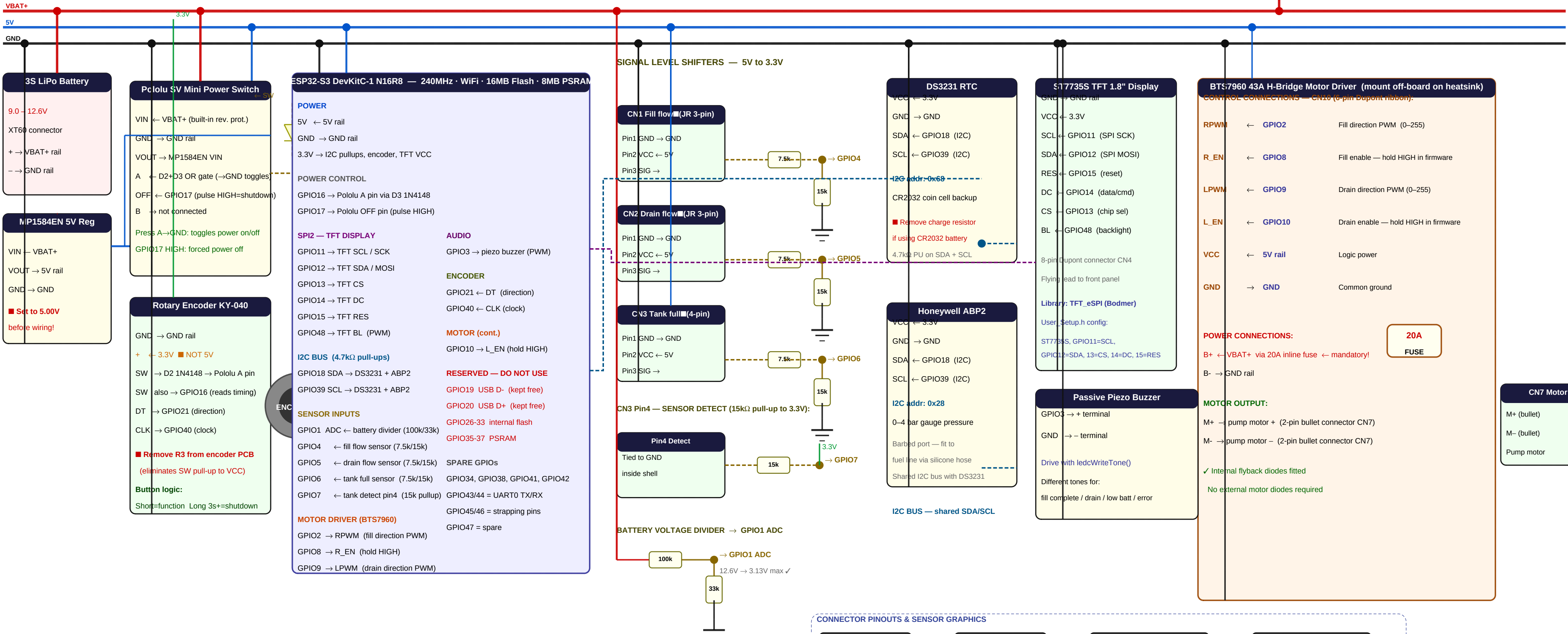


MCP Fuel Station v2 — Circuit Diagram

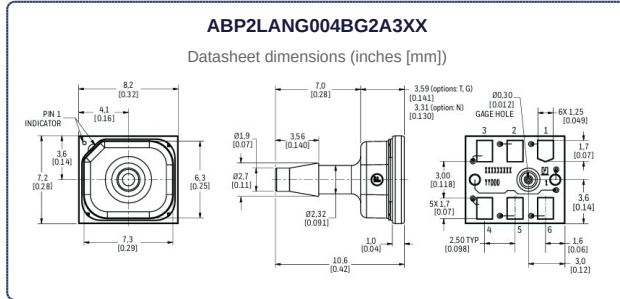
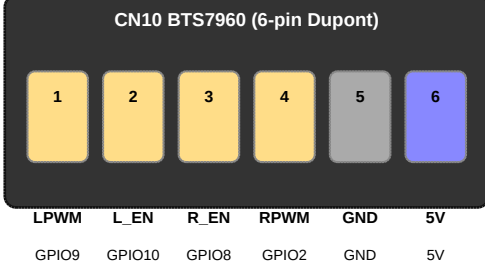
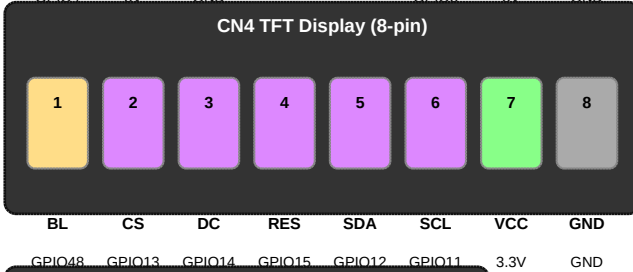
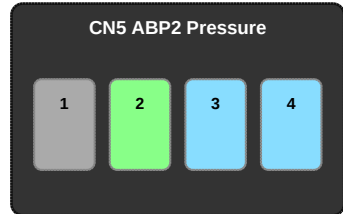
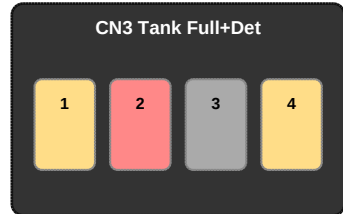
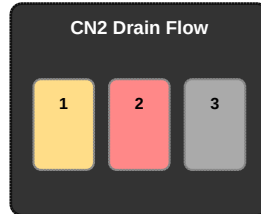
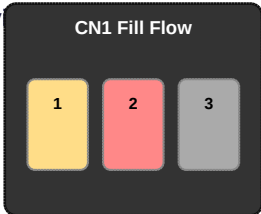
ESP32-S3 N16R8 · 3S LiPo · MP1584EN 5V · Pololu SV Mini · BTS7960 · ST7735S TFT · DS3231 RTC · Honeywell ABP2 · KY-040 Encoder

SIGNAL TYPES: VBAT+ (9–12.6V) 5V regulated 3.3V GND Signal / GPIO SPI bus I2C bus



CONNECTOR PINOUTS (JR style — left to right)

CONNECTOR PINOUTS & SENSOR GRAPHICS



Pins: 1=NC 2=VCC(3.3V) 3=NC 4=GND 5=SDA(GPIO18) 6=SCL(GPIO39)

BUILD NOTES

- Pre-set MP1584EN to exactly 5.00V with multimeter BEFORE connecting to board.
- Remove R3 from encoder PCB — eliminates SW pull-up that can hold Pololu A pin HIGH.
- Remove DS3231 charge resistor (next to 1N4148 on module back) before fitting CR2032.
- BTS7960 must be mounted off-board on aluminium heatsink — it gets hot under load.
- 20A inline fuse is mandatory on VBAT+ wire to BTS7960 B+ terminal.
- Test all power rails with multimeter before inserting ESP32.
- Encoder: power from 3.3V only — NOT 5V. This avoids 5V signals on ESP32 GPIO pins.
- I2C addresses: DS3231 = 0x68 · ABP2 = 0x28 · No conflicts.
- TFT_eSPI library: configure User_Setup.h with ST7735S driver, GPIO11=SCL, 12=SDA, 13=CS, 14=DC, 15=RES.
- Diode OR gate: SW → D2 → A pin, GPIO16 → D3 → A pin. Pressing SW or GPIO16 LOW toggles Pololu. GPIO17 HIGH = OFF pin.
- Short press: firmware reads GPIO16 timing for function select. Long press 3s+: firmware pulses GPIO17 HIGH.

MCP Fuel Station v2

Circuit Diagram Rev 2.0

ESP32-S3 · Single board

A2 Landscape